

Career Interest Recommender System

A Python-Based Solution for Student Career Guidance

Project Report

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Executive Summary

The Career Interest Recommender is a Python console application designed to address a critical challenge faced by students, i.e identifying which technology career domain aligns with their interests and aptitudes. Through a simple yet effective rule-based scoring system, the application analyzes student responses to six targeted questions and provides personalized career recommendations across six major technology domains: AI/ML, Web Development, Cybersecurity, Data Science, UI/UX Design, and Cloud & DevOps. The project demonstrates practical application of fundamental Python concepts while delivering real-world value to students navigating early career decisions.

Introduction

Students entering their first year of college often face uncertainty about which technology specialization to pursue. This project develops a Python-based career interest recommender that helps students identify suitable tech domains through a simple questionnaire and rule-based scoring system.

Objectives

- Develop a console-based Python application using fundamental programming concepts
 - Implement a rule-based scoring algorithm mapping interests to career domains
 - Provide actionable output including study plans and future career roles
-

System Design

The application follows a three-phase architecture: Input (collect responses), Processing (weighted scoring), and Output (recommendations with study plans). The scoring algorithm awards 2 points for primary domain matches and 1 point for related skills. Each of the 6 questions targets specific career characteristics, differentiating between domains effectively.

Implementation and Results

Sample Output

Your Recommended Career Path is:

-> Cloud & Devops

Interest level: Medium - This area matches some of your interests.

Career Path: Cloud & Devops

What you should study first:

- Basics of Linux and command line
- Intro to cloud platforms (AWS, Azure, GCP)
- Version control (Git) and CI/CD basics

Future career options:

- Cloud Engineer
- Devops Engineer
- Site Reliability Engineer

Note: This is a simple guide. Explore online courses and resources for deeper learning

Discussion

Strengths: Uses fundamental Python concepts, addresses real student needs, no dependencies.

Limitations: Rule-based approach, limited depth, no data persistence. Future enhancements could include file storage.

Conclusion

The Career Interest Recommender successfully demonstrates how fundamental Python programming can solve real-world problems affecting students. By combining simple data structures, conditional logic, and user input handling, the application delivers personalized career guidance in an accessible format.

The project achieves its core objectives: it requires only basic Python knowledge, and provides actionable recommendations that students can immediately apply to their academic planning. The rule-based scoring system, while simple, effectively differentiates between diverse technology domains and accounts for overlapping skill sets.

Beyond its functional utility, the project serves as an excellent learning vehicle for students transitioning from theoretical programming concepts to practical application development. The clear correlation between code structure and real-world problem-solving reinforces fundamental programming principles while building student confidence.

Appendix A: Complete Source Code

```
print("==== Career Interest Recommender ====")

print()

print("Answer the following questions with: 1 (Yes) or 0 (No)")

print()
```

Scores for each career path

```
scores = {

    "AI & Machine Learning": 0,
```

```
"Web Development": 0,  
"Cybersecurity": 0,  
"Data Science": 0,  
"UI/UX Design": 0,  
"Cloud & DevOps": 0  
}
```

Extra info: what to study + future options

```
study_plan = {  
  "AI & Machine Learning": {  
    "study": [  
      "Python basics and libraries like NumPy, Pandas",  
      "Math: linear algebra, probability, statistics",  
      "Intro to machine learning concepts"],  
    "Future Occupation": [  
      "Machine Learning Engineer",  
      "Data Scientist",  
      "AI Engineer"]  
  },  
  "Web Development": {  
    "study": [  
      "HTML, CSS, basic JavaScript",  
      "Python with Flask or Django",  
      "Responsive web design basics"],  
    "Future Occupation": [  
      "Frontend Developer",  
      "Backend Developer",  
      "Full Stack Web Developer"]  
  },  
  "Cybersecurity": {  
    "study": [  
      "Computer networks and OS basics",
```

```
"Linux commands and system security basics",
"Common attacks (phishing, malware, SQL injection)"],
"Future Occupation": [
"Cybersecurity Analyst",
"Ethical Hacker",
"Security Engineer"]
},
"Data Science": {
"study": [
"Python, statistics, and data analysis",
"Pandas, NumPy, and data visualization",
"Basic machine learning algorithms"],
"Future Occupation": [
"Data Analyst",
"Data Scientist",
"Business Intelligence Engineer"]
},
"UI/UX Design": {
"study": [
"Design principles and color theory",
"Wire framing tools (Figma, Adobe XD basics)",
"User research and usability testing"],
"Future Occupation": [
"UI Designer",
"UX Designer",
"Product Designer"]
},
"Cloud & Devops": {
"study": [
"Basics of Linux and command line",
"Intro to cloud platforms (AWS, Azure, GCP)",
```

"Version control (Git) and CI/CD basics"],

"Future Occupation": [

"Cloud Engineer",

"Devops Engineer",

"Site Reliability Engineer"]}]}

Helper: get 0/1 safely

```
def ask_yes_no(question):
```

```
    while True:
```

```
        ans = input(question + " ")
```

```
        if ans in ["0", "1"]:
```

```
            return int(ans)
```

```
    print("Please enter only 1 for Yes or 0 for No.")
```

6 short questions

```
q1 = ask_yes_no("Do you enjoy solving logical or math problems? (1/0)")
```

```
if q1==1:
```

```
    scores["AI & Machine Learning"] += 2
```

```
    scores["Data Science"] += 2
```

```
    scores["Cybersecurity"] += 1
```

```
q2 = ask_yes_no("Do you like creating or designing visually appealing things? (1/0)")
```

```
if q2==1:
```

```
    scores["UI/UX Design"] += 2
```

```
    scores["Web Development"] += 1
```

```
q3 = ask_yes_no("Are you curious about how websites and apps are built? (1/0)")
```

```
if q3==1:
```

```
    scores["Web Development"] += 2
```

```
    scores["Cloud & Devops"] += 1
```

```
q4 = ask_yes_no("Do you like finding patterns in data or charts? (1/0)")
```

```
if q4==1:
```

```
scores["Data Science"] += 2
scores["AI & Machine Learning"] += 1
```

```
q5 = ask_yes_no("Are you interested in protecting systems from hackers? (1/0)")
if q5==1:
    scores["Cybersecurity"] += 2
    scores["Cloud & Devops"] += 1
```

```
q6 = ask_yes_no("Do you enjoy working with tools, automation, or cloud platforms? (1/0)")
if q6==1:
    scores["Cloud & Devops"] += 2
    scores["AI & Machine Learning"] += 1
```

Find best path(s)

```
max_score = max(scores.values())
best_paths = [name for name, val in scores.items() if val == max_score]
```

Interest level

```
if max_score <= 2:
    interest_level = "Low – You are still exploring different areas."
elif max_score <= 4:
    interest_level = "Medium – This area matches some of your interests."
else:
    interest_level = "High – This area strongly matches your interests!"
```

```
print("\n===== Result =====\n")
```

```
if len(best_paths)==1:
    print("Your Recommended Career Path is:")
else:
    print("You match well with multiple career paths:")
```

```
for path in best_paths:
```

```

print("->", path)

print("\nInterest level:", interest_level)

print("\nScores for reference:")
for name, val in scores.items():
    print(f"{name}:{val}")

```

Show what to study and future options for each best path

```

print("\n===== What You Should Study & Future Options =====\n")

for path in best_paths:
    info = study_plan[path]
    print("Career Path:", path)
    print("What you should study first:")
    for item in info["study"]:
        print(" -", item)
    print("Future career options:")
    for job in info["Future Occupation"]:
        print(" -", job)
    print() # blank line between paths

```

```

print("Note: This is only a simple guide.")

```

Appendix B: Usage Instructions

Save the code as `careerrecommender.py`, open terminal, navigate to the file directory, and execute `python careerrecommender.py`. Answer each question with 1 (Yes) or 0 (No), then review recommendations.

Requirements: Python 3.6+ on any operating system, no additional libraries needed